CS330 Final Project Reflection

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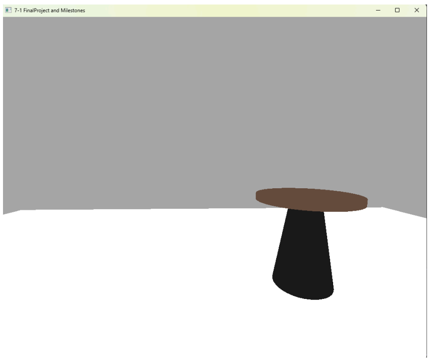
The scene I chose is a modern styled wooden armchair, with a beige fabric covered cushion, a wooden floor, grey rug and a small side table that is made from a matte black colored metal, a wooden top, which has a blue covered book resting on top.

Justify development choices for your 3D scene

After going through the project examples attached in the “Copy of More Project Examples” announcement, I especially liked the image of the throne shown in the top left:

I had the idea of doing something similar, but in a modern style, using an armchair instead of a throne. I often joke that an armchair is the modern throne, so I felt that a scene which includes an elegant armchair, a table and a book can achieve a good balance between being fun and being highly achievable.

I started my scene by simply building the table as I thought that would be the easiest to do

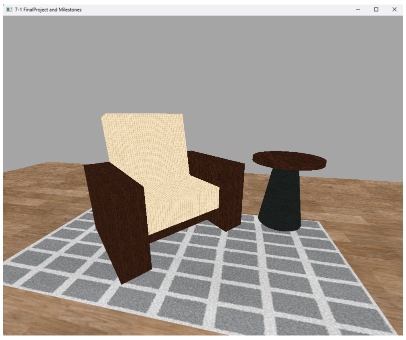


I also changed the background to be a similar grey to the walls in my image so that I could differentiate between the floor and the background more easily.

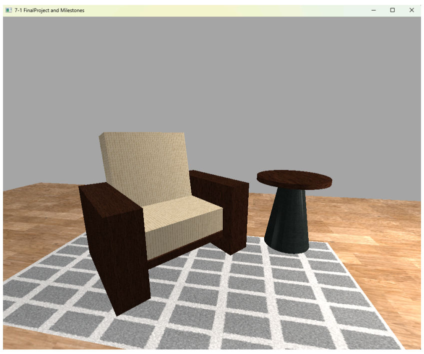
After I gained some confidence I decided to tackle the most complex object in my scene, the armchair. This part was more challenging. I decided to build the armchair assuming a 40 degrees rotation so that it will be facing a similar direction as in the chosen image. After that I added colors to match the image.


The next task was to add textures. Since we had to use copyright free images I found the website [ambientCG](). I adjusted the textures UVScale to apply tiling according to the XYZ scale for the mesh wherever possible to make sure that the textures won't stretch, but in some places the textures looked better and more inline with the image if I gave them different values, for example the floor looked better when it was indeed stretched to some degree. My main focus was that my scene would be as similar to the image as possible, which meant that I'd place whichever values I found to be the most similar to the scene after playing around with them.

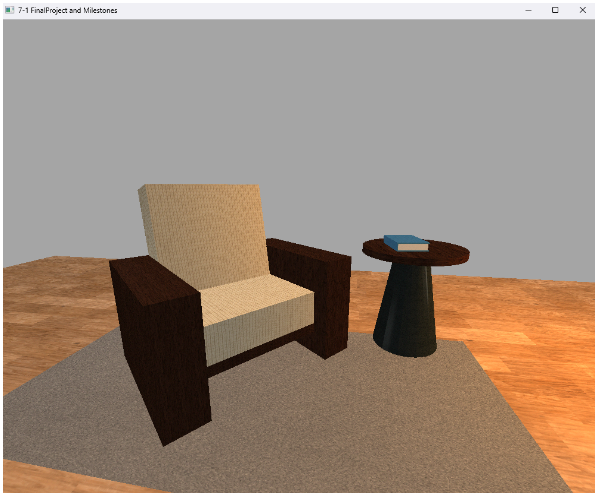
I couldn’t find a grey rug texture so I used a checkered rug texture. I also couldn’t find a texture with the correct color I wanted for the cushions so I edited a texture I found to change its color to beige which did the trick really well.



When it came to lights I first used a directional light to make sure my scene isn’t completely dark. I noticed that the light in my chosen image mostly came from the right-front side, so I then added a point light right there. And a weaker fill light on the other side to keep the scene balanced.



When it came to the final project submission, I noticed a requirement for having a colored light, so I changed the light to a cozy orange. I added a book to rest on the table, and decided to edit the rug texture to be grey to make it look more similar to the original image:



Overall I'm very satisfied with the result. I think that it looks cozy and replicates my scene of choice well. I have to say it was satisfying to see the gradual improvement which is why I took so many screenshots along the way.

Explain how a user can navigate your 3D scene

Well, since the directional movement with the WASD keys and the mouse navigation were already implemented, I only had to add the E & Q keys to pan up and down, O & P keys to toggle between orthographic and perspective projections and hook up the mouse scroll wheel to control the movement speed.

It was mostly straightforward. I looked at how the WASD keys were already implemented as well as the camera.h file to implement the E & Q keys. For the O key, I set the camera to snap in front of the scene parallel to the floor and for the P key I've set the camera to snap back to the starting position as defined in the constructor. For the mouse scroll movement speed I created a custom function "Mouse_Scroll_MovementSpeed_Callback", where I mapped the scroll wheel to increment and decrement the movement speed, but I made sure to set a boundary with 2 basic if statements to enforce that the camera movement speed can't go below 1 or above 50.

Explain the custom functions in your program that you are using to make your code more modular and organized

To make sure my code is modular and organized, other than creating custom functions throughout my code for the new functionality added, such as the "Mouse_Scroll_MovementSpeed_Callback" function mentioned above, I refactored my code from my previous submissions and created a helper function named "RenderShape" which accepts the parameters for scale, position, rotation, texture tag, texture scale, material and shape type.

By utilizing this reusable helper method I made sure my code has a higher level of abstraction.

It significantly reduced the number of lines of code I originally had and made my project submission more organized and easier to read.